ACS-3922 Fall 2026

Assignment 3

Due Date: Wednesday, November 18 at 11:59 pm

Part A – Complete Tutorial

Complete Tutorial 4 so that you have a complete Top Down Shooter to use as a basis for Part B. Include the Camera Controller provided in Appendix C of the tutorial.

Part B – Build a Game

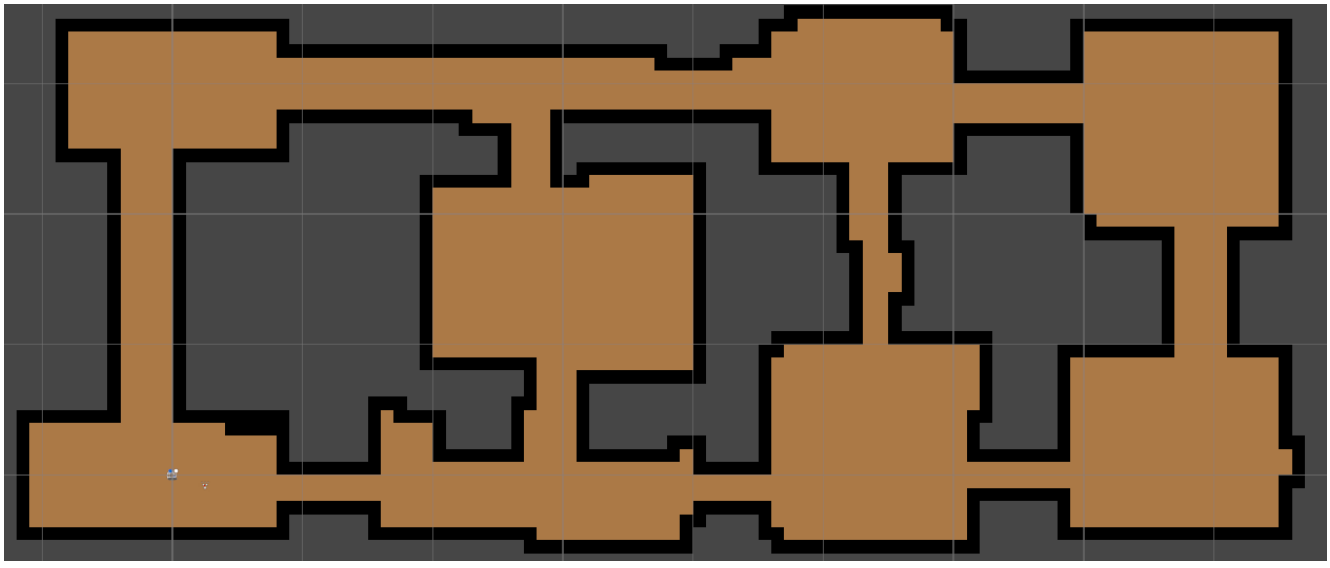
Using what you built in Tutorial 4 as the basis, create a game based on the design and requirements below.

Design:

For this design, you will create a basic dungeon crawler that uses our top down shooter framework for combat.

Requirements:

- Use a tilemap to generate walls and floor
- You must design a map of rooms and corridors for the player to explore.
- Your dungeon must occupy a space at least 57 tiles wide x 30 tiles high.
    - Your map doesn't need to fill every square inch of this area (i.e. you can have larger rooms connected by smaller corridors), but the foot print of the entire map should at least cover the area.



An example of an acceptable map layout.

- You must implement an exit that the player must find in order to win the game
- You must have at least 15 total enemy instances in your map
- You must implement a health system for the player
    - Player has a maximum amount of health
    - Health is subtracted from the player when they take damage
    - When the player's health drops below 1, they are killed and the game is over.
- You must implement a hazard that damages the player when they walk over it
    - The hazard must have a visual element that identifies it as a hazard
    - The hazard should damage the player when they first step onto the hazard
    - The hazard should then damage the player after a set time interval if they continue to stand on or walk over the hazard
- You must implement a health pickup that replenishes the player's health.
    - The pickup must have a visual element that identifies it as a healing item
- You must have destructable crates that have a 15% chance of dropping health items when they are destroyed.
- You must implement a HUD that displays the player's current health
    - This can be either number based (i.e. Health: 5) or icon based (i.e. display 5 heart icons and remove them as damage is applied.)
- You must have a game over screen
    - The game over screen must have a unique message for when the player looses (is killed)
    - The game over screen must have a unique message for when the player wins (escapes)

Part C – Game Feel

Create a new markdown file called A3PartC.md.

1) Be sure to cite any sources that you may have used for your game.
2) Give your game a title and a story: this can be very simple. A couple of sentences that establish: the world of the game, who the main character is, and the goal of the game.
3) Write a paragraph on what Game Feel you wanted to achieve with this game. Explain how the game feel you were designing for relates to your narrative or theme.
4) Write a paragraph on what you did in your game to achieve the Game Feel you were aiming for. Identify what parts of input, response, context, polish, and metaphor you manipulated, and how you manipulated them in order to try and achieve your desired game feel.

Assignment 3 – Side Quest

I wanted to provide everyone with an opportunity to improve their midterm grades if they were unhappy with the score they received. The Side Quest is that opportunity. The Side Quest is **100% optional.** Taking part in the Side Quest will **not** affect your grade for Assignment 3.

If you complete the Side Quest, then you can earn up to 10 percentage points on your midterm grade. The bonus amount you'll receive will be the mark you receive on Assignment 3 applied to that 10 percentage points. For example If you get 80% on Assignment 3 and you complete the Side Quest, you'll get +8 percentage points added to your midterm mark. If you earn 50%, you'll get +5. Note: Your mid-term mark cannot exceed 100%.

To complete the Side Quest you must first complete Assignment 3. Then you have to complete the additional requirements laid out below in Side Quest Part A and Side Quest Part B. The requirements are **not graded** and **do not affect your mark for Assignment 3**. They are treated as a pass/fail requirements for achieving the Side Quest. Either you complete them all and achieve the Side Quest, or you don't. **Failure to satisy any of the Additional Requirements means you fail the Side Quest.**

Side Quest Part A – Additional Requirements

Requirements:

- Exit locations must be dynamically choosen at runtime (i.e. they cannot be placed in a single static location on the map)
    - The exit must spawn somewhere on the opposite side of the map relative to the player's spawn point
    - HINT: One way to achieve this is to have a list of static positions the exit could potentially spawn at and pick one at random
    - If you use a list of static positions to pick where to spawn your exit, there must be at least 5 different potential spawning points where the exit could show up
- You must implement a gold system to track the amount of gold the player is carrying
    - The player starts each new game with zero gold
- You must implement a gold pickup that adds to the player's gold total
    - The pickup must have a visual element that identifies it as a gold item
- You dungeon must contain destructible crates that have a 25% chance of dropping a gold pickup when they are destroyed
    - These can be the same destructible crates that have a 15% chance of dropping health pickups, or you can have two distinct types of crates
- You must implement a hazard that stuns the player when they walk over it
    - The hazard should stun the player for a set duration when they first step onto it
    - The hazard should then stun the player again after a set time interval if they continue to stand on or walk over the hazard

    - The interval between the first and subsequent stuns must be at least double the duration of the stun itself
    - The hazard deals no damage to the player, only applies a stun
    - The hazard must be visually distinct from the damage hazard implemented in Part B of the assignment
- You must implement stun feedback on the player
    - While the player is stunned but not damaged, you must display some visual feedback
    - This visual feedback must be distinct from the damage feedback (i.e. a different colour, or made from particles instead of sprite colour swapping, etc.)
- You must implement a dash ability on the player
    - When the Right Mouse Button is pressed, the player should dash in the current direction of movement
    - If the dash button is pressed when the player is not moving, nothing should happen
    - Dashing must have a 2 second cool-down between dashes
    - While on cool-down, dash cannot be triggered and clicking the right mouse button should do nothing
    - HINT: One way of implementing dashing is to treat it like a knock-back triggered by the player input and using the current input direction
- You must have at least 3 different types of enemies in your dungeons (tutorial enemy counts as 1 type so you must implement 2 new types)
- An enemy counts as a new type if and only if:
    - It is visually distinct from other types of enemies
    - It has a unique behaviour or unique combination of behaviours that distinguishes it from other types of enemies
    - It has attributes that distinguish it from other types of enemies
- You must add 2 additional levels to your dungeon (3 floors total)
- The dungeon created for Assignment 3 is your first floor
- The second floor of your dungeon must occupy a space at least 76 tiles wide x 40 tiles high
- Your second dungeon floor must have at least 20 total enemy instances
- The third floor of your dungeon must occupy a space at least 19 tiles wide x 10 tiles high
- The third floor must have at least 1 boss enemy instance
- A enemy counts as a boss type if and only if:
    - It is visually distinct from other types of enemies
    - It has a unique behaviour or unique combination of behaviours that distinguishes it from other types of enemies
    - It has attributes that distinguish it from other types of enemies
    - By virtue of its distinguishing attributes and behaviours, it presents a larger challenge to defeat than other enemy types
- Your third floor exit does not need to be dynamically placed (i.e. it can always spawn in the same location), but it must not spawn until the boss is defeated

- Your HUD must display the players current gold total
- Your HUD must display the current floor
- Your game over screen must display the total collected gold when the player wins
    - It does not need to display the total collected gold when the player looses

Side Quest Part B – Log Your Side Quest

Create a new markdown file called A3SideQuestPartB.md.

1) Identify and name your 3 enemy types. Explain how their unique behaviour or unique combination of behaviours distinguishes them from each other. Explain how their attributes distinguish them from each other.
2) Identify and name your boss enemy type. Explain how their unique behaviour or unique combination of behaviours distinguishes them from the other enemy types. Explain how their attributes distinguish them from the other enemy types. Explain how they present a larger challenge to defeat than other enemy types.

Good luck and happy questing!